

# Week 5 – Workshop

## Isostasy: Ocean–Continent Structure & Isostatic Gravity



Geophysics 3A: Potential Fields & Geothermics

July 25, 2026

**Duration:** 3 hours

**Setting:** Workshop room (small-group whiteboard work)

### Preparation

**Pre-reading:** Course Notes, Ch. 7

**Lecture videos:**

**Notes:**

### Purpose

This workshop develops a quantitative understanding of isostatic equilibrium and its gravity expression by combining column-based reasoning, ocean–continent structure, and slab gravity approximations. Students work from a single general mass-balance principle throughout, applying it first to a crust-only ocean–continent comparison, then to a layered lithosphere, then to gravity. Airy and Pratt are introduced as named patterns within this balance, not as separate starting formulas.

### Learning Outcomes

By the end of this workshop, you should be able to:

- Explain isostatic equilibrium as equal lithostatic pressure (or mass per unit area) at a compensation depth, for columns of arbitrary structure.
- Diagnose when a model result (e.g. an unrealistic density) indicates a missing physical component, and propose what is missing.
- Recognize whether a given compensation pattern is consistent with the Airy or Pratt end-member, and explain why.
- Derive the isostatic gravity correction from mass balance and the infinite slab formula, and explain why large topography does not necessarily produce large gravity anomalies.
- Explain compensation as a spectrum rather than a binary state, and use misfit between predicted and observed gravity to reason about the degree of compensation.

### Session Flow (180 min)

#### 0–10 min: Warm-Up: Icebergs

Isostasy is a state of equilibrium related to the buoyancy of the lithosphere over a geophysically fluid mantle. In this exercise, we'll examine the buoyancy of icebergs and discuss the similarity to the lithosphere that we'll be studying.

#### 10–20 min: Mini-Lecture: The General Isostatic Balance

- Isostatic equilibrium as pressure balance at a common compensation depth

- Handling columns of unequal height

The full statement of this balance, and how to apply it to columns of unequal height, is given in the theory box at the start of the next activity.

### **20–50 min: Activity 5.A: Oceanic vs. Continental Crust, Part 1**

Each group balances an oceanic and a continental column, assuming compensation occurs solely through crustal thickness, and solves for the oceanic crust density required for equilibrium. See the activity sheet for the theory and setup.

### **50–60 min: Mini-Lecture: End-Member Terminology**

- Naming the two patterns you just used: Airy (uniform density, variable thickness) and Pratt (uniform thickness, variable density)
- Schematic comparison of the two patterns, and why real structure usually mixes both

### **60–85 min: Activity 5.B: Oceanic vs. Continental Crust, Part 2**

The model is extended to include a lithospheric mantle beneath both columns. Groups rebalance the columns, estimate the density contrast between oceanic and continental lithospheric mantle, and relate it to mineralogy (e.g. garnet content). See the activity sheet.

### **85–125 min: Activity 5.B: Isostatic Gravity Correction**

Working from the same balance, groups derive the compensating root thickness, apply the infinite slab formula (Week 4) to topography and root, and show that their gravity effects cancel under full compensation. See the activity sheet for the full derivation sequence.

### **125–155 min: Numerical Demonstration**

The instructor builds a lithospheric cross section in `isostasy_gui.py` and computes its gravity response under varying degrees of compensation and layered structure. Groups work through the accompanying activity sheet as the demonstration proceeds.

### **155–170 min: Wrap-up**

Each group prepares a brief summary addressing:

- One case in this workshop where an unrealistic result told you something was missing from the model, and what that turned out to be.
- Whether a compensation pattern you examined today was closer to Airy, Pratt, or a mix, and why.
- One reason a compensated load can produce little or no gravity anomaly, and one reason a real gravity residual might not be zero.
- One observation (e.g. seismology or petrology) that could help distinguish between isostatic models that fit the same gravity data equally well.

### **170–180 min: Preview: Magnetism**

Next week we move from gravity to magnetism. Like gravity, magnetic methods respond to a physical property contrast—in this case, magnetic susceptibility rather than density. However, magnetism differs in important ways:

- magnetic susceptibility is controlled by trace rather than bulk mineralogy;
- rock magnetization is a vector (not a scalar);

- the response depends on the direction of the regional field; and
- the source contribution changes sign with polarity.

We will use hands-on demonstrations with magnets and rock samples to build intuition before examining how crustal structure is recorded in magnetic anomalies.

## **Workshop Notes**

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## Activity 5.A

### Oceanic vs. Continental Crust

#### Purpose

This activity applies the principle of isostatic equilibrium to oceanic and continental lithosphere. By comparing increasingly realistic column models, you will investigate why simple crust-only models fail and explore the role of lithospheric mantle structure and composition in supporting long-wavelength topography.

#### Learning Outcomes

By the end of this activity, you should be able to:

- Set up and solve isostatic balance equations for two-layer lithosphere columns.
- Explain the limitations of applying simple Airy-style reasoning to ocean–continent structure.
- Describe how mantle lithosphere composition (particularly garnet fraction) contributes to the density contrast between oceanic and continental mantle.
- Discuss the relationship between lithosphere composition, density, and long-term isostatic stability.

#### Theory: The General Isostatic Balance

Isostatic equilibrium requires that pressure be equal at a common *depth of compensation* beneath any two neighboring columns, regardless of how the mass within each column is distributed between layers. If a column is built from  $N$  layers of thickness  $h_i$  and density  $\rho_i$ , the pressure it exerts at its base is

$$P = g \sum_{i=1}^N \rho_i h_i.$$

Because  $g$  is common to every column, isostatic equilibrium between two columns reduces to equality of their layer-integrated mass per unit area:

$$\sum_{i=1}^N \rho_i h_i \Big|_{\text{Column 1}} = \sum_{i=1}^N \rho_i h_i \Big|_{\text{Column 2}}.$$

**Handling unequal column heights.** Real columns rarely have the same total height — one may stand higher (positive elevation) or lower (e.g. beneath an ocean) than the other. To compare them at a common depth, extend the *shorter* column downward with additional material of mantle density  $\rho_m$  until both columns reach the same base level, the depth of compensation (Figure 1). This filler thickness is not a separate physical layer — it is simply the bookkeeping needed to align two columns of different height at a common depth. Written this way, elevation is not a separate term added to the balance; it is already accounted for as part of one column's total thickness.

*Note:* If you are not sure in advance which column is higher, it does not matter which one you assign the elevation to — assign it either way and solve. Placing it on the correct column gives a positive elevation; placing it on the other column simply gives the same result with a negative sign.

**What you will do in Part A.** You are given the layer thicknesses and densities for a continental and an oceanic column. One unknown density is missing. Write the pressure balance above for these two specific columns, extend the shorter column to the depth of compensation using mantle density, and solve for the missing density directly.

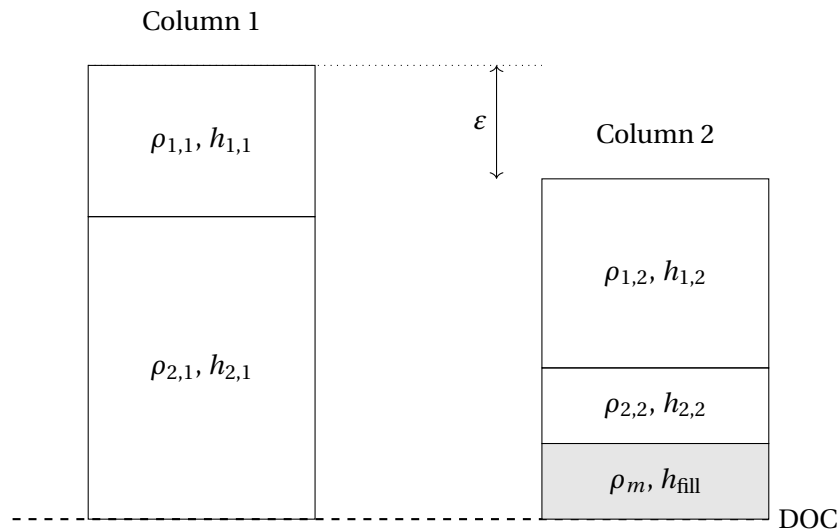


Figure 1: Two generic lithospheric columns of different total height, aligned at a common depth of compensation (DOC). Column 2 stands lower by elevation  $\epsilon$ ; this is accounted for by an additional mantle-density filler thickness  $h_{fill}$  at its base, *not* by adding a separate  $\epsilon$  term to the pressure balance.

## Part 1: Testing a Crust-only Isostasy Model

### Given:

**oceanic crust thickness:**  $h_{oc} = 7 \text{ km}$

**ocean depth:**  $h_w = 3.7 \text{ km}$

**seawater density:**  $\rho_w = 1030 \text{ kg m}^{-3}$

**continental crust thickness:**  $h_{cc} = 40 \text{ km}$

**continental crust density:**  $\rho_{cc} = 2850 \text{ kg m}^{-3}$

**continental elevation:**  $\epsilon = 0.8 \text{ km}$

**mantle density:**  $\rho_a = 3330 \text{ kg m}^{-3}$

**Instructions:** Assume isostatic compensation occurs solely through crustal thickness. Compute the density of the oceanic crust,  $\rho_{oc}$ , that balances the oceanic and continental columns. Compare your result with typical oceanic crust densities ( $\sim 2900\text{--}3000 \text{ kg m}^{-3}$ ) and discuss why the crust-only assumption may fail.

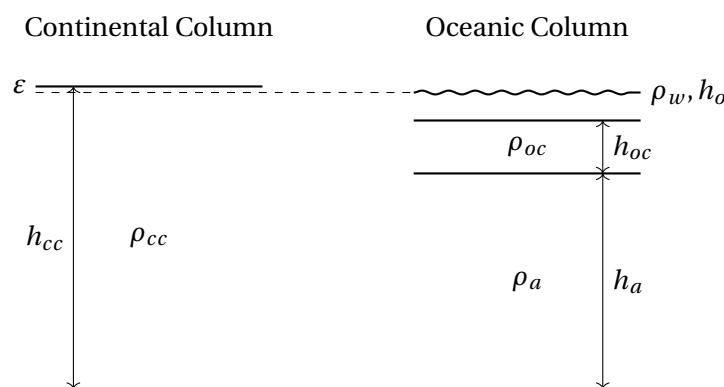


Figure 2: Comparison of isostatic columns for the continental and oceanic crust for Part 1.

### Question 1. One-layered isostasy.

Compute the required oceanic crust density  $\rho_{oc}$ .

**Question 2.** Reflection.

Compare with expected oceanic crust densities and discuss why the simple crust-only model may not yield realistic values.

**Part 2: Two-layer Lithosphere Model**

**Given additional information:**

**Oceanic lithospheric mantle thickness**  $h_{om} = 75 \text{ km}$

**Oceanic lithospheric mantle density:**  $\rho_{om} = 3330 \text{ kg m}^{-3}$

**Continental lithospheric mantle thickness:**  $h_{cm} = 150 \text{ km}$

**Continental lithospheric mantle density:**  $\rho_{cm} = ?$

**Asthenosphere density:**  $\rho_a = 3320 \text{ kg m}^{-3}$

**Instructions:** Let's add a two-layered lithosphere using average lithospheric thicknesses for the oceanic and continental settings. Our goal will be to see what we can learn about the relative density difference between continental and oceanic mantle lithosphere. In the process, we'll show how including the mantle lithosphere resolves the unrealistic crust-only result.

Table 1: Modal proportions of minerals in the oceanic and continental lithospheres.

| Mineral       | Oceanic | Continental |
|---------------|---------|-------------|
| olivine       | 75      | 60          |
| orthopyroxene | 25      | 17          |
| clinopyroxene | 10      | 11          |
| garnet        | 0       | 12          |

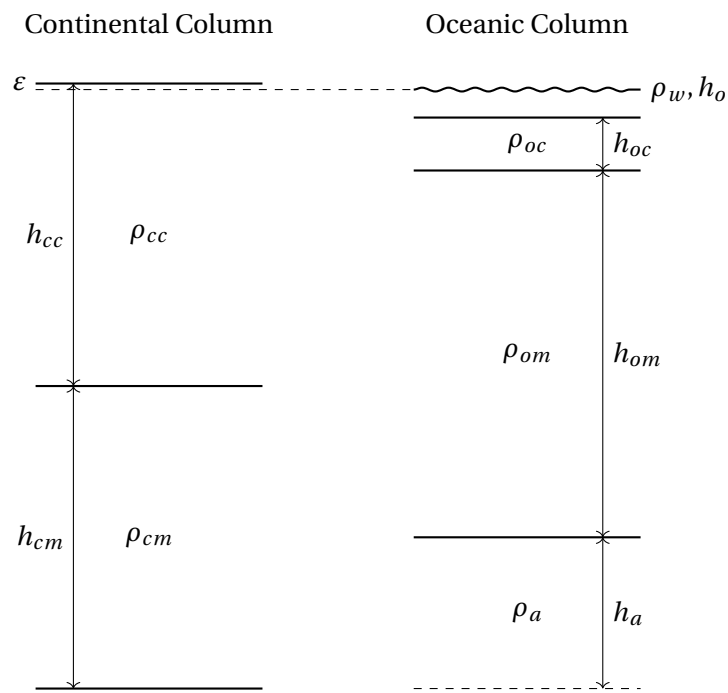


Figure 3: Comparison of isostatic columns for the continental and oceanic crust for Part 2.

Oceanic lithospheric mantle is assumed to be composed of depleted peridotite with a density of  $\rho_{om} = 3330 \text{ kg m}^{-3}$ . Continental lithospheric mantle may contain a higher proportion of garnet.

**Density of depleted mantle (peridotite):**  $\rho_d = 3330 \text{ kg m}^{-3}$

**Density of garnet:**  $\rho_g = 3600 \text{ kg m}^{-3}$

**Question 3.** Two-layered isostasy.

Setup the isostatic equations for a two-layered mantle lithosphere. If we assume an average oceanic crustal density  $\rho_{oc} = 2900\text{--}3000 \text{ kg m}^{-3}$ , what density of the continental mantle lithosphere?

**Question 4.** Explain.

Explain how including the mantle lithosphere resolves the unrealistic density obtained in the

crust-only case.

**Question 5.** Discuss.

Discuss the role of garnet fraction in the continental lithospheric mantle for the density contrast. Compute the percentage difference garnet between the oceanic and continental lithosphere and compare your result with the modal proportions in Table 1.

## Key Ideas

**An unrealistic parameter value is diagnostic, not just wrong.** When solving a balance forces a physical property outside its plausible range (as with the required oceanic crust density here), that is a signal about what the model is missing, not merely an error to note and move past. The next step is always to ask what layer, mechanism, or assumption was left out.

**Density contrast, not absolute density, drives isostatic balance.** What matters in every column comparison is the difference between adjacent layers' densities, not their individual values. This is why even modest compositional differences between two otherwise similar layers can carry real isostatic consequences.

**Buoyancy depends on composition as much as thickness.** A thick layer is not automatically a heavy one — its contribution to the balance depends on what it is made of. Any two properties that trade off in a balance (here, thickness and density) mean you cannot interpret one without also considering the other.

**Adding a layer resolves one unknown but introduces others.** Moving from a crust-only to a two-layer model produced a more realistic result, but only by introducing a new density that itself had to be constrained. Extending a model to fix an implausible result is rarely free — it is a trade of one assumption for another, better-motivated one.

## Activity 5.B

### Isostatic Gravity Correction

#### Purpose

In this problem, you will derive the isostatic gravity correction under the assumption of isostasy, starting from a Bouguer slab formulation and progressing to a practical implementation using topography and a reference crustal thickness.

#### Learning Outcomes

By the end of this activity, you should be able to:

- Set up and solve a mass-balance equation between a reference column and a topographic column to obtain the compensating root thickness.
- Apply the infinite slab formula (Week 4) to compute the gravity effect of surface topography and its compensating root.
- Show algebraically that the topography and root contributions cancel under perfect compensation (i.e., no flexure).
- Recognize whether a compensation pattern is consistent with one of the isostatic end-members.
- Explain why the isostatic correction is model-dependent and discuss how incomplete compensation or flexural rigidity modifies the result.

#### Background

Consider a reference Earth model with a constant crustal thickness  $T_0$ . Surface topography  $h(x)$  is compensated by a crustal root  $r(x)$  such that columns of equal width have equal mass above a compensation depth. This is the same isostatic balance from Activity 5.A, applied here to a single topographic column compared against a flat reference column.

Let:

- $\rho_c$  = crustal density
- $\rho_m$  = mantle density
- $h(x)$  = surface elevation
- $r(x)$  = additional root thickness relative to  $T_0$

#### Geometry

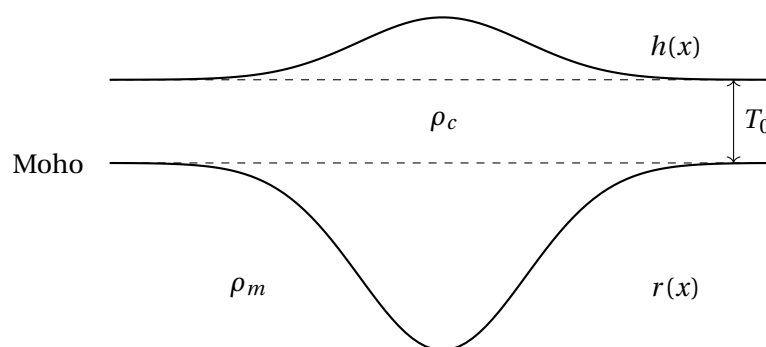


Figure 4: A mountain with a topographic expression  $h(x)$  and compensating root  $r(x)$ .

## Tasks

### Question 1. Part 1: Root Thickness

- i Using the general mass-balance principle from Activity 5.A, write the pressure balance between the reference column ( $T_0, \rho_c$ , mantle below) and the topographic column ( $T_0 + h + r, \rho_c$ ), evaluated at the depth of compensation. Show that this reduces to

$$\rho_c h = (\rho_m - \rho_c) r.$$

- ii Solve for the root thickness  $r(x)$  in terms of  $h(x)$ .

- iii Is this compensation pattern consistent with one of the end-member models, Airy or Pratt, and if so why?

### Question 2. Part 2: Gravity of an Infinite Slab

Recall the infinite slab formula from Week 4: the gravitational attraction of an infinite horizontal slab of thickness  $t$  and density  $\rho$  is

$$g = 2\pi G \rho t.$$

- i Write an expression for the gravitational effect of the topography.

- ii Write an expression for the gravitational effect of the crustal root. Carefully define the density contrast.

**Question 3.** Part 3: Substitution and Simplification

- i Substitute your expression for  $r(x)$  into the root gravity term.

- ii Show that the total gravity effect of the topography plus its compensating root simplifies to:

$$g_{\text{total}} = 0$$

**Question 4.** Part 4: Practical Isostatic Correction

In practice, crustal thickness is not known everywhere. Instead, we assume a reference thickness  $T_0$  and compute  $r(x)$  from observed topography.

- i Explain why the isostatic correction is model-dependent.

- ii Write the final expression for the isostatic correction in terms of  $h(x)$  only.  
You should show that the gravity effect of the compensating root is:

$$g_{\text{root}} = -2\pi G\rho_c h$$

and that, under perfect compensation, the combined effect of topography and root cancels exactly.

### Question 5. Extension (Optional)

1. If a load is only partially compensated, does the topography–root cancellation from Part 3 still hold exactly? What would you expect to see in the residual gravity anomaly, and what does its sign tell you?

2. Partial compensation of this kind arises physically from lithospheric flexure. Why does a rigid lithosphere leave short-wavelength loads less compensated than long-wavelength ones?

### Key Ideas

**Under perfect compensation, topography and its root cancel exactly.** The gravity signal from a mountain ( $+2\pi G\rho_c h$ ) is exactly cancelled by its deeper crustal root ( $-2\pi G\rho_c h$ ). A perfectly compensated terrain

therefore has no isostatic gravity anomaly — any residual signal indicates incomplete compensation or lateral density variations.

**This crustal-thickness-only pattern is an end-member.** Constant crustal density with a compensating root is exactly the pattern you classified in Part 1 — the same general balance would look different for a case with different fundamental assumptions.

**The isostatic correction depends on a model, not just data.** To apply the correction in practice we must assume a crustal density and a reference thickness  $T_0$ ; both are uncertain. Different choices of these parameters produce different corrections, so the corrected gravity inherits model uncertainty.

**Flexural rigidity distributes the compensation laterally.** A rigid lithosphere can support loads over a finite wavelength range without immediate local isostatic adjustment. Short-wavelength topography is not compensated; long-wavelength topography is. The effective elastic thickness of the lithosphere controls where this transition occurs.

## Activity 5.C

### Numerical Demonstration: Isostatic Gravity Along a Cross Section

#### Purpose

In this demonstration, the instructor will use `isostasy_gui.py` to build a lithospheric cross section along a chosen topographic profile, compute its gravity response under varying degrees of compensation and layered lithospheric structure, and compare the predicted gravity to observations. The purpose is to see the single-column reasoning from Activities 5.A and 5.B applied continuously along a real profile, and to meet the one genuinely new idea of the session: that compensation is not all-or-nothing.

#### Learning Outcomes

By the end of this demonstration, you should be able to:

- Explain how the column-by-column mass balance from Activity 5.A extends to a continuous topographic profile.
- Describe what the degree of compensation  $C$  represents, and predict how changing it should affect the amplitude of the predicted gravity anomaly.
- Explain how the choice of layered lithospheric structure (crust only vs. crust + mantle lithosphere) affects the gravity prediction for the same topography.
- Use misfit between predicted and observed gravity to reason about how well (and where) a profile is locally compensated.
- Identify what a computational model needs beyond what was required for the single-column calculations in 5.A and 5.B.

#### Background

**From columns to a profile.** Everything in this demonstration rests on the same balance you used in Activity 5.A: at the depth of compensation, column-integrated mass must match between neighboring columns. The only difference here is scale — instead of comparing two columns by hand, the program applies this balance (or a partial version of it, below) at every point along a chosen cross section, built from a loaded crustal thickness model rather than two fixed values.

**Degree of compensation.** So far, every problem has assumed *full* local compensation: the root exactly satisfies the mass balance from 5.A. In reality, compensation is often partial, since lithospheric strength allows some of a load to be supported elastically rather than buoyantly (Activity 5.B, Extension). We describe this with a single parameter  $C$ , where

$$r(x) = C r_{\text{Airy}}(x), \quad 0 \leq C \leq 1,$$

and  $r_{\text{Airy}}(x)$  is the fully compensated root thickness from Activity 5.B.  $C = 1$  recovers full local compensation;  $C = 0$  means the load is entirely supported elastically, with no compensating root at all.

**Gravity of a real profile.** The infinite slab formula from 5.B assumed topography with no lateral edges. A real profile does not, so the program instead sums the gravity effect of many finite blocks along the cross section — the same discrete 2-D prism approach you saw in Week 3 (Activity 3.B, Q11), not a new formula.

#### App Overview

- Select two points to define a cross section across the loaded topography.

- Load a crustal thickness model for the same profile.
- Specify a layered lithospheric model (layer densities and thicknesses), following the same structure as Activity 5.A Part B.
- Set the degree of compensation  $C$  and compute the predicted gravity anomaly along the profile.
- Compare the predicted gravity to observed gravity along the same profile, if loaded.

## Tasks

Follow along as the instructor builds a cross section and adjusts each parameter in turn. Answer the questions below as you go — they are written to apply to whichever profile is selected.

### Question 1. Full compensation baseline

With  $C = 1$ , does the predicted gravity anomaly broadly track or oppose the topography along the profile? Explain why, referring back to what you showed in Activity 5.B, Part 3.

### Question 2. Varying the degree of compensation

As the instructor decreases  $C$  from 1 toward 0, predict first, then observe: does the amplitude of the predicted gravity anomaly increase or decrease? Explain in terms of how much compensating root mass remains at each value of  $C$ .

### Question 3. Finding the best-fit compensation

If observed gravity is available for this profile, adjust  $C$  to find the value that best matches the predicted and observed gravity. What does that value of  $C$  tell you about how this section of lithosphere is actually being supported?

### Question 4. Model dependence

With  $C$  held fixed, the instructor will change the assumed crustal density and/or reference lithospheric thickness. How does the predicted gravity change? Relate this to the "model-

dependent correction" Key Idea from Activity 5.B.

**Question 5.** Adding a layer

The instructor will now add or modify a mantle lithosphere layer, as in Activity 5.A Part B. Does the degree of compensation needed to fit the observed gravity change once this layer is included? Why might a two-layer model change your interpretation compared to a crust-only model?

**Question 6.** Uniform compensation?

Does a single value of  $C$  fit the observed gravity equally well across the whole profile, or does the fit degrade in some places (e.g. near a sharp topographic feature or a continent–ocean boundary)? What would that tell you about assuming one compensation state for an entire region?

**Question 7.** What changed, computationally

Compared to the single-column, by-hand calculations in 5.A and 5.B, what additional ingredients did this model need to predict gravity along a real profile?

## Key Ideas

**Compensation is a spectrum, not a switch.** Local isostatic balance (as in 5.A and 5.B) and fully elastic support are two ends of a continuum; most real topography is compensated to some intermediate degree, and that degree can be estimated by comparing predicted and observed gravity.

**A single column-balance principle scales up without changing.** The same mass balance used for two columns by hand applies, unchanged, at every point along a continuous profile — modelling a whole cross section is a matter of scale and bookkeeping, not new physics.

**Fitting a model is itself an inference about the Earth.** The value of  $C$  (and the layered structure) that best matches observed gravity is not just a curve-fitting exercise — it is an estimate of how a real piece of lithosphere is mechanically supported, with the same model-dependence caveats that applied to the isostatic correction in 5.B.